Aga Khan University Examination Board

Notes from E-Marking Centre on SSC-I Chemistry Examination April/ May 2019

Introduction

This document has been produced for the teachers and candidates of Secondary School Certificate (SSC) Part I Chemistry. It contains comments on candidates' responses to the 2019 SSC-I Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations

In comparison to previous years, overall performance of the candidates has improved. However, there is still room for improvement. Mentioned below are few concepts on which teachers need to focus and give candidates more drill and practice to have a strong grip.

a. Difference between the terms characteristic and formation of compounds.
b. Basic components of chromatographic techniques and their purpose.
c. Scientific reasons behind the periodicity of different properties of elements.
d. Concept of isotopes.
e. Relationship amongst vapour pressure, intermolecular forces and boiling points of liquids.
f. Scientific reason for using gold, silver and platinum in daily life.
g. Use of formulas in different stepwise calculations of empirical and molecular formula of compounds.
h. Difference between diffusion and effusion in gases.
i. Oxidation-reduction reaction in rusting of iron along with the components that accelerate/ inhibit rusting process.

Detailed Comments:

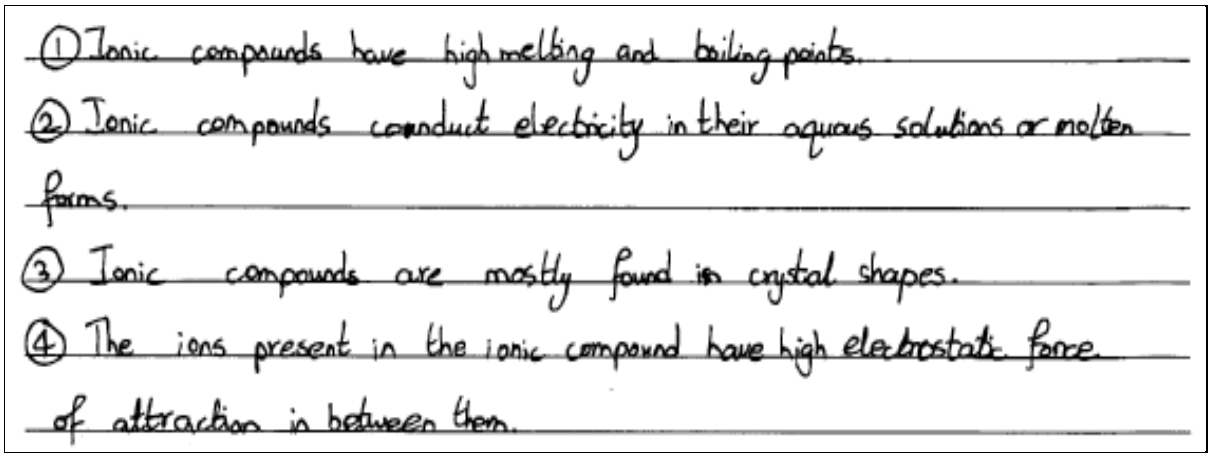
Constructed Response Questions (CRQs)

Question 1:

State any FOUR characteristics of an ionic compound.

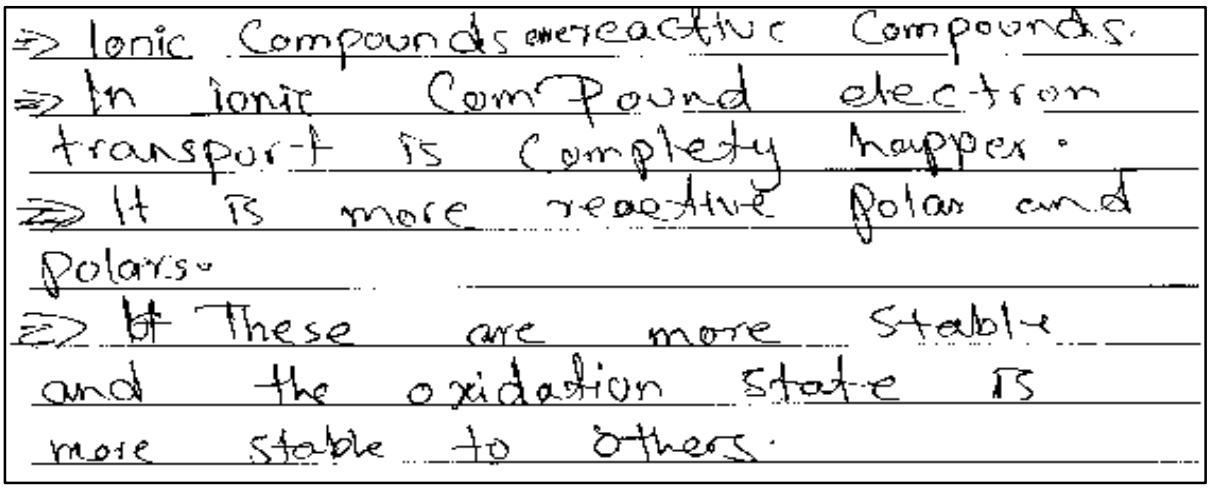
Better responses showed the correct characteristics of ionic compounds. Candidates demonstrated clear understanding of physical properties of ionic compounds including their solubility in water, 3-dimensional crystalline structure, high melting and boiling point, electrical conductance in molten or solution state along with the chemical characteristics like polarity, presence of ions, and strong electrostatic force of attraction between ions.

Example:



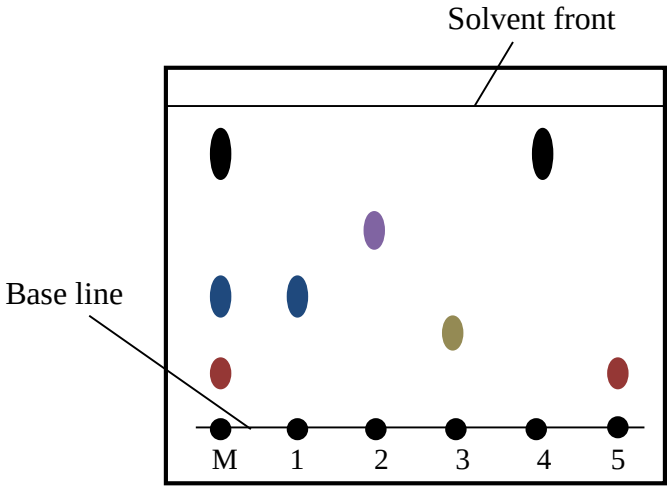
Weaker responses demonstrated poor conceptual knowledge about the characteristics of ionic compounds. Most of these responses described the formation of ionic bond by complete transfer of electrons but failed to figure out the characteristics. Mostly weaker responses could not determine the difference between high melting point and sharp/ fixed melting points. Candidates described about metals by mentioning the presence of free electrons, malleability and ductility. Similarly, these responses mentioned the presence of electrical conductance in ionic compounds because of having free electrons, instead of mentioning the conditions such as molten or solution form. A few of the weaker responses also mentioned the high reactivity of ionic compounds without mentioning molten of solution form.

Example:



Question 2:

An analytical chemist uses chromatography to investigate the types of amino acids present in a mixture of amino acids. He represents the mixture as M, and the known amino acids as 1, 2, 3, 4 and 5. During experimentation, he ensures that the base line containing spots is placed above the solvent level. After experimentation, he sprays the chromatogram with a solution of ninhydrin. The given chromatogram shows his investigation result.

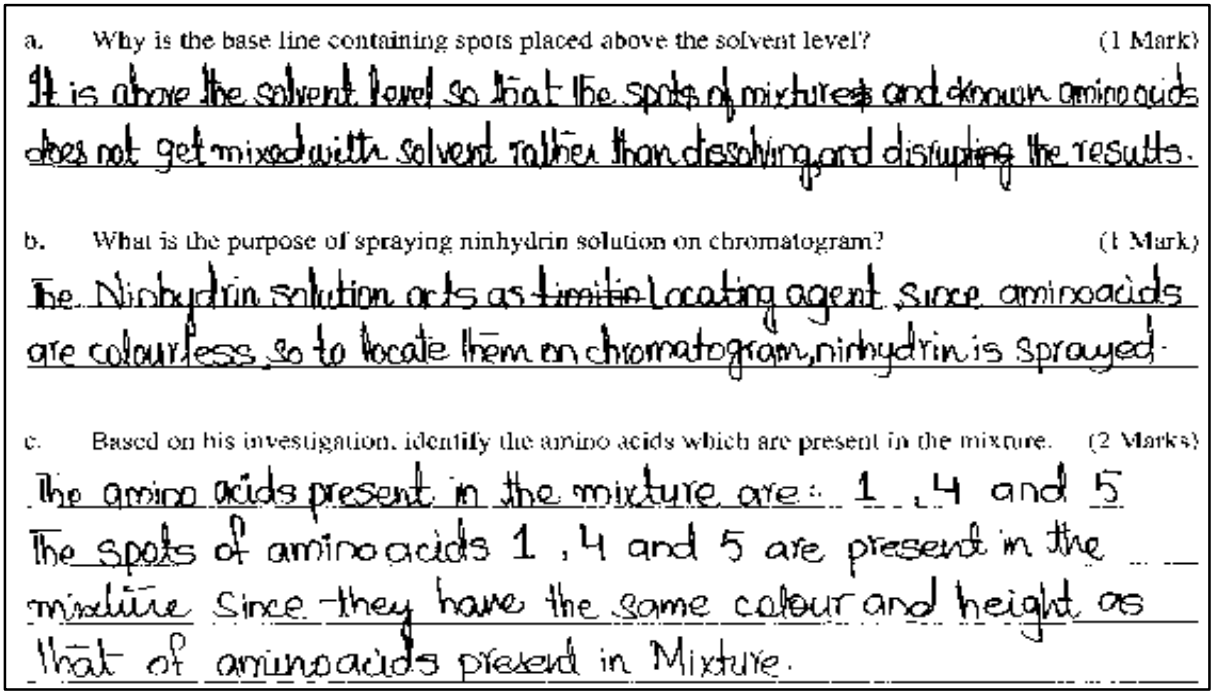


Chromatogram

a. Why is the base line containing spots placed above the solvent level?
b. What is the purpose of spraying ninhydrin solution on chromatogram?
c. Based on his investigation, identify the amino acids which are present in the mixture.

Better responses showed clear understanding of the concept of chromatography. Candidates were able to explain the reason for the placement of the base line above the solvent level in part 'a', i.e. to prevent mixing of spots with the solvent. In part 'b', candidates correctly mentioned the purpose of ninhydrin spray is to locate/ visualise the spots on the chromatogram. Moreover, in part 'c', candidates successfully identified the composition of the mixture by representing the correct numbers of amino acids.

Example:



a. Why is the base line containing spots placed above the solvent level? (1 Mark)

It is above the solvent level so that the spots of mixtures and known amino acids does not get mixed with solvent rather than dissolving and disrupting the results.

b. What is the purpose of spraying ninhydrin solution on chromatogram? (1 Mark)

The Ninhydrin solution acts as ~~a~~ locating agent since amino acids are colourless so to locate them on chromatogram, ninhydrin is sprayed.

c. Based on his investigation, identify the amino acids which are present in the mixture. (2 Marks)

The amino acids present in the mixture are: 1, 4 and 5
The spots of amino acids 1, 4 and 5 are present in the mixture since they have the same colour and height as that of amino acids present in Mixture.

Weaker responses depicted lack of conceptual understanding regarding chromatography. In part 'a', candidates wrongly stated the reason for the placement of baseline above the solvent level, i.e. for the measurement of spots/ for separation of colour dyes/ to prevent chromatogram from getting wet etc. Whereas, in part 'b', candidates showed lack of knowledge regarding locating agents. These responses included statements like ninhydrin spray is used to find the result/ to investigate the result/ to stop the movement of spots/ to check the amino acids in mixture. Most of the candidates in weaker responses wrongly identified the numbers of amino acids. In part 'c', a few of the weaker responses also represented the composition of mixture with names of different amino acids like Glysine/ functional groups/ elements that are present in amino acids rather than their correct numbers.

Example:

The line containing spots Placed above the solvent after measuring.

b. What is the purpose of spraying ninhydrin solution on chromatogram? (1 Mark)

when we spray ninhydrin solution on chromatogram it kills the germs and bectoria.

c. Based on his investigation, identify the amino acids which are present in the mixture. (2 Marks)

-Based on his investigation, there are 8 amino acids which are present in the mixture.

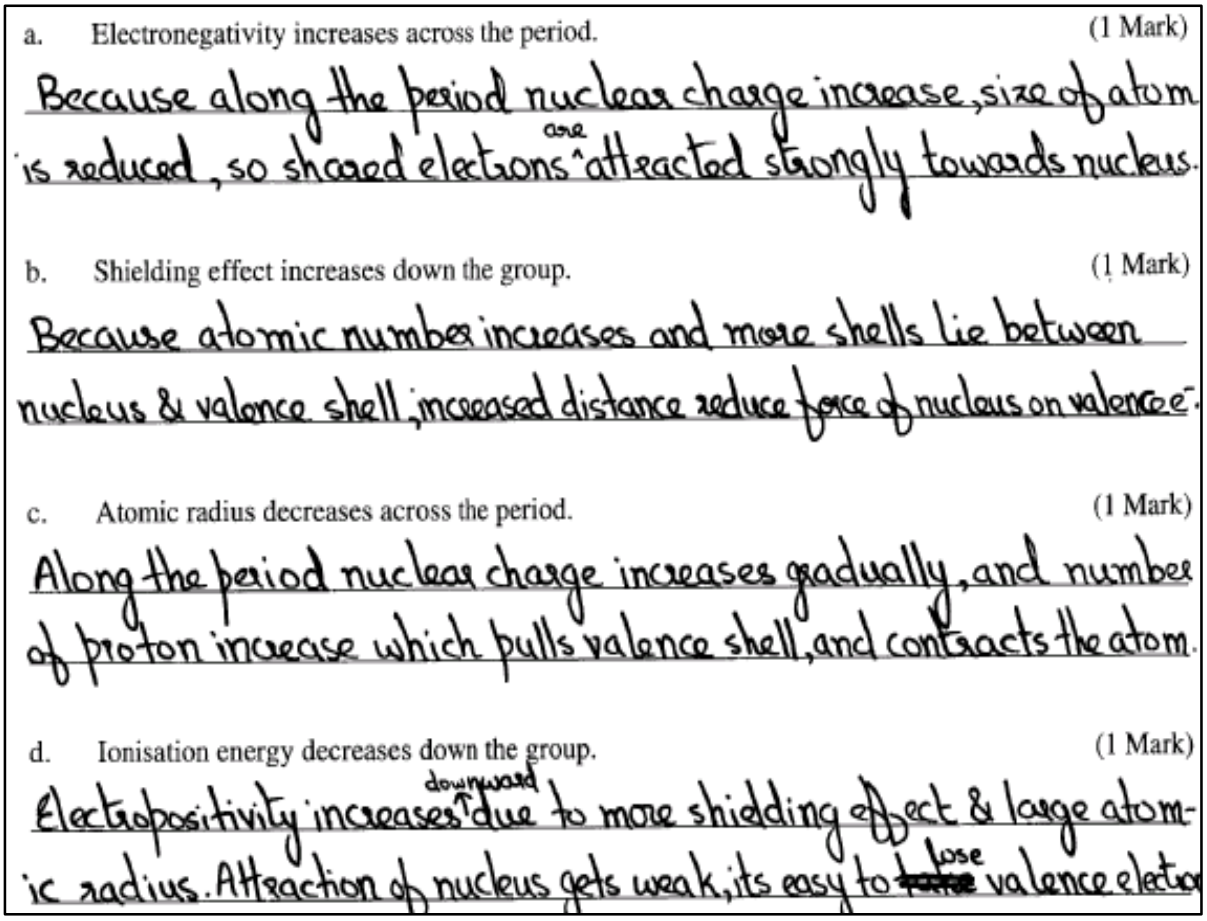
Question 3:

Write a reason for each of the following periodic trends observed in the periodic table.

a. Electronegativity increases across the period.
b. Shielding effect increases down the group.
c. Atomic radius decreases across the period.
d. Ionisation energy decreases down the group.

Better responses depicted the correct reasoning for the given periodic trends. For example, in part 'a', increase in electronegativity across the period is due to increase in number of charges on the nucleus that attracts the bonding pair of electrons more strongly/ effective nuclear charge shortens the distance of the shared pair of electrons from the nucleus; thus, increasing the power to attract the shared pair of electrons. Similarly, in part 'b', candidates mentioned Shielding effect increases down the group because with the increase of atomic number, the number of electrons in an atom also increases that results in an increase of shielding effect. In part 'c, atomic radius decreases across the period because as we go across a period, electrons are added to the same energy level. At the same time, protons are being added to the nucleus. The concentration of more protons in the nucleus creates a higher effective nuclear charge. In other words, there is a stronger force of attraction pulling the electrons closer to the nucleus resulting in a smaller atomic radius. Furthermore, candidates in part 'd' had clarity that ionisation energy decreases down the group because electrons are farther from the nucleus and thus, it is easier to remove the outermost electron/ shielding effect increases down the group, i.e. the inner electrons at lower energy levels essentially block the protons' force of attraction toward the nucleus. It therefore, becomes easier to remove the outer electron.

Example:



a. Electronegativity increases across the period. (1 Mark)

Because along the period nuclear charge increase, size of atom is reduced, so shared electrons are attracted strongly towards nucleus.

b. Shielding effect increases down the group. (1 Mark)

Because atomic number increases and more shells lie between nucleus & valence shell, increased distance reduce force of nucleus on valence e⁻.

c. Atomic radius decreases across the period. (1 Mark)

Along the period nuclear charge increases gradually, and number of proton increase which pulls valence shell, and contracts the atom.

d. Ionisation energy decreases down the group. (1 Mark)

Electropositivity increases downward due to more shielding effect & large atomic radius. Attraction of nucleus gets weak, its easy to lose valence electron

Weaker responses showed lack of critical thinking and reasoning skills. In most of the weaker responses, candidates connected all the properties with atomic size but failed to explain the reason behind increasing and decreasing atomic size. Most of the responses showed incomplete reasons for all four trends. Multiple responses had stated the same reasons for part 'a' and 'c' as these were asked about across the period. Similarly, in part 'b' and 'd', candidates had stated the same reasons for the two different properties as these were asked about down the group.

Example:

Q.3. (Total 4 Marks)

Write a reason for each of the following periodic trends observed in the periodic table.

a. Electronegativity increases across the period. (1 Mark)

It is so because there is an increase in the proton number (atomic number), which increases electronegativity.

b. Shielding effect increases down the group. (1 Mark)

It is so because there is an addition of shell as we move down and Zeff also decreases.

c. Atomic radius decreases across the period. (1 Mark)

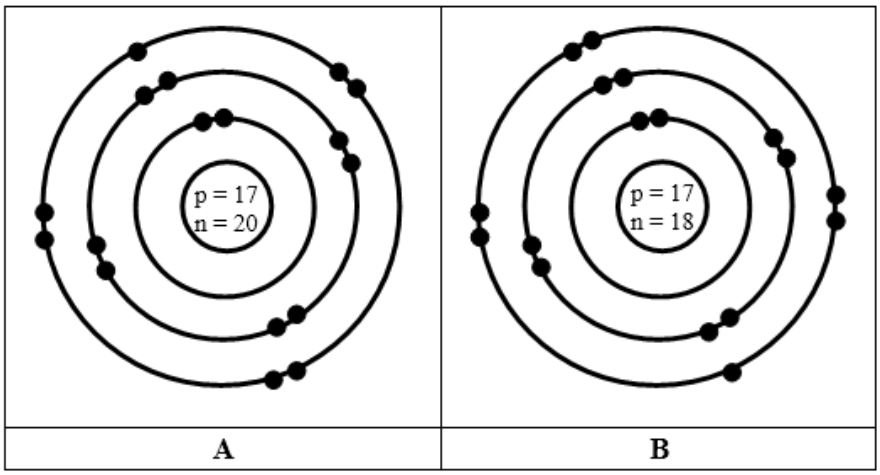
It is so because there is no addition of shell in the period only the atomic number increases.

d. Ionisation energy decreases down the group. (1 Mark)

It is so because ionization energy depend upon the valence electron and valence electron do not increases down a group.

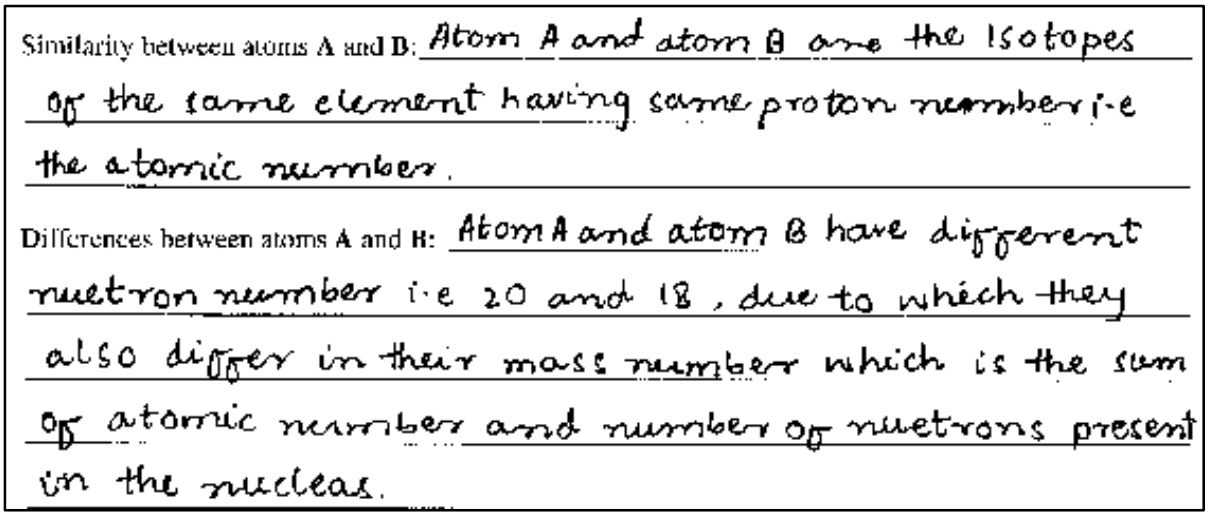
Question 4:

Write any ONE similarity and TWO differences between the given atoms **A** and **B**.



Better responses showed the clarity of concept regarding isotopes. In many responses, candidates identified the given atoms A and B as isotopes of each other/ isotopes of chlorine atom. Beside that these responses distinctively mentioned the basic similarities between isotopes, i.e. same atomic number/ same proton number/ same number of electrons/ same number of valence electrons/ same electronic configuration/ same number of electrons are required in valence shell/ both can form anion by accepting one electron. As for the differences between the given isotopes, candidates mentioned that both the atoms have different mass number and the number of neutrons.

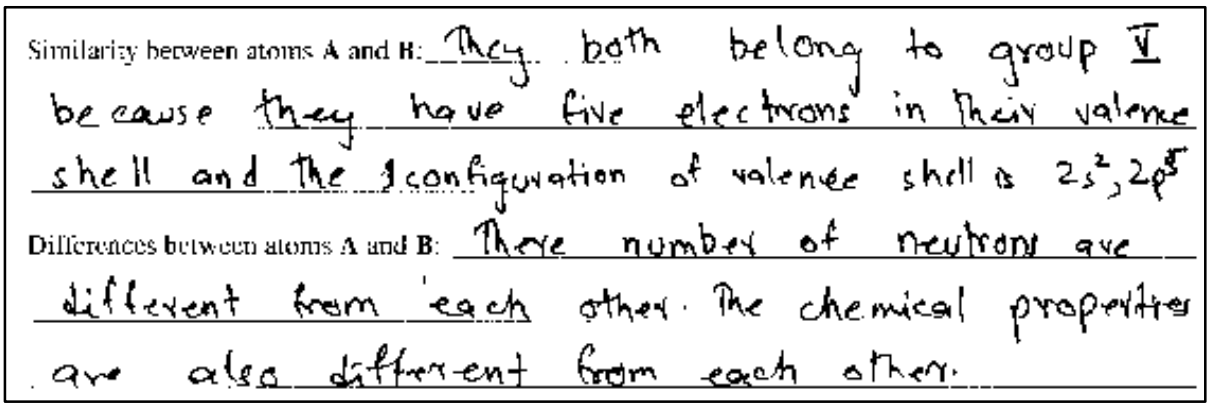
Example:


Similarity between atoms A and B: Atom A and atom B are the isotopes of the same element having same proton number i.e the atomic number.

Differences between atoms A and B: Atom A and atom B have different neutron number i.e 20 and 18, due to which they also differ in their mass number which is the sum of atomic number and number of neutrons present in the nucleas.

Weaker responses depicted a lack of recognition of the isotopes of atoms. Most of the candidates comprehend diagram A as an atom of element and diagram B as its isotope. Both having the same number of shells was another misconception that was frequently encountered. Furthermore, candidates demonstrated that both atoms have different electronic arrangement/ different size/ different chemical properties/ belong to same group of the periodic table. Most of the weaker responses, somehow, were able to identify the different number of neutrons successfully, but failed to write the second dissimilarity.

Example:


Similarity between atoms A and B: They both belong to group V because they have five electrons in their valence shell and the configuration of valence shell is $2s^2, 2p^5$

Differences between atoms A and B: There number of neutrons are different from each other. The chemical properties are also different from each other.

Question 5:

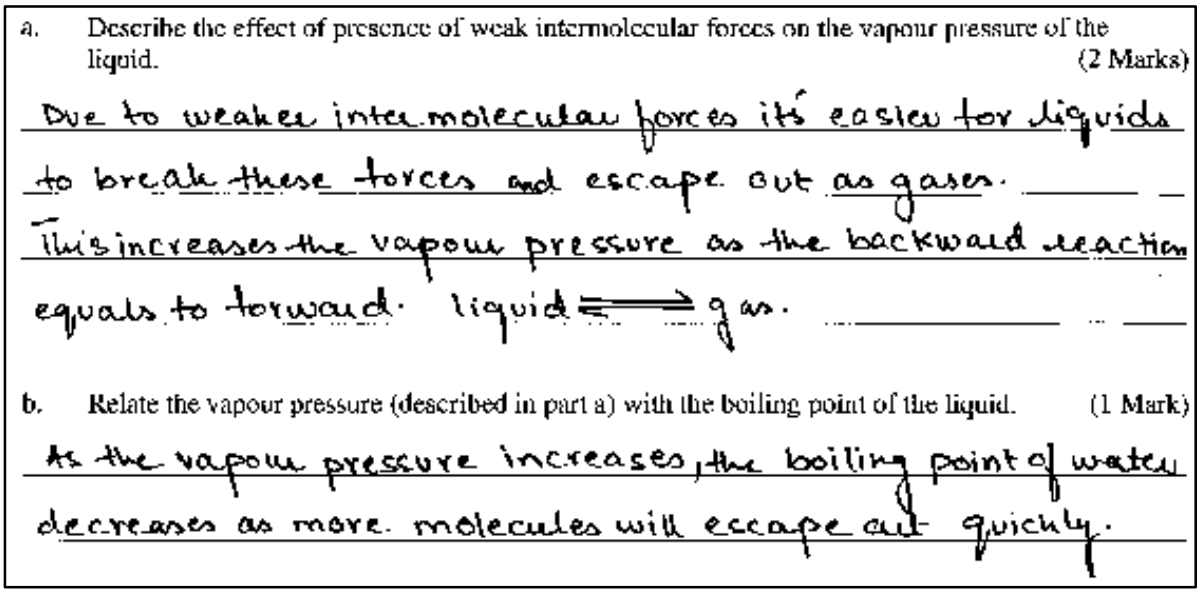
Consider a closed container with liquid having weak intermolecular forces. This liquid experiences both, vapour pressure and boiling point at constant temperature.

a. Describe the effect of presence of weak intermolecular forces on the vapour pressure of the liquid.

b. Relate the vapour pressure (described in part a) with the boiling point of the liquid.

Better responses in part 'a', correctly interpreted that the presence of weak intermolecular forces in liquids will be responsible for the greater rate of vapourisation. This is because the molecule can easily break their interactions and are likely to be converted into gas that, in turn, will increase the movement of molecules and ultimately the vapour pressure. Whereas, in part 'b', such responses showed clarity in conceptual understanding of the inverse relationship between vapour pressure and boiling point of the liquids having weak intermolecular forces.

Example:


a. Describe the effect of presence of weak intermolecular forces on the vapour pressure of the liquid. (2 Marks)

Due to weaker inter-molecular forces it's easier for liquids to break these forces and escape out as gases. This increases the vapour pressure as the backward reaction equals to forward. liquid ⇌ gas.

b. Relate the vapour pressure (described in part a) with the boiling point of the liquid. (1 Mark)

As the vapour pressure increases, the boiling point of water decreases as more molecules will escape out quickly.

Weaker responses displayed theoretic inadequacy of the whole context. In part 'a', candidates failed to relate the presence of weak intermolecular forces with that of vapour pressure of the liquid. These responses showed misconceptions such as it is the vapour pressure which weakens the intermolecular forces. In part 'b', candidates could not clarify the relation between greater vapour pressures and lowering the boiling points in liquids.

Example:

a. Describe the effect of presence of weak intermolecular forces on the vapour pressure of the liquid. (2 Marks)

The vapour pressure will weaken the intermolecular forces more and the molecules will scatter away, occupying more volume with high vapour pressure.

b. Relate the vapour pressure (described in part a) with the boiling point of the liquid. (1 Mark)

Vapour pressure also increases the boiling point of liquid as liquids will loose their specific positions of molecules

Question 6:

Give a scientific reason why gold is

a. used in making jewellery.
b. alloyed with copper when making jewellery.
c. extensively used in dentistry.

Better responses furnished the question with scientific reasoning. These responses showed the clarity of the concept regarding properties of metals and their usage. In these responses, candidates correctly mentioned in part 'a' that gold is used in jewellery making because of its malleability/ non-reactive nature as it is a noble metal. In part 'b', candidates explained that copper is alloyed with gold to make it hard. Whereas, in part 'c', candidates distinctively state the extensive use of gold in dentistry is resistance to corrosion/ non-allergic as the reasons.

Example:

a. used in making jewellery. (1 Mark)

because gold is an inert metal, thus it is not affected by air or any mineral and it is also shiny in appearance

b. alloyed with copper when making jewellery. (1 Mark)

because gold is very soft, therefore it is alloyed with copper to make it harder and more malleable

c. extensively used in dentistry. (1 Mark)

because it is malleable and resistent to corrosion. Mostly compound containing gold are used in dentistry.

Weaker responses indicated that the candidates were aware of the properties of metals but failed to manipulate correctly in the specifically asked usage. Candidates, in part 'a', wrote some non-scientific/ irrelevant reasons like gold is used in jewellery making because it is an expensive metal/ a good conductor/ good reflector of light so it does not require polish. In part 'b', candidates mostly showed emphasis on heat conduction because of copper. Moreover, these responses justified the alloying of gold with copper so as to prevent it from corrosion/ make gold cheaper in cost. Whereas, in part 'c', candidates showed unfamiliarity with the terms 'dentistry' and 'extensively'. Again in this part, candidates mentioned the reason of the shiny appearance of gold.

Example:

a. used in making jewellery. (1 Mark)

Gold is used in jewellery to make the jewellery shinny and attractive and has melallic luster

b. alloyed with copper when making jewellery. (1 Mark)

Copper is used in jewellery to prevent it from rusting alloyed when gold alloyed with copper and also for etc electroplating to give new look to the jewellery.

c. extensively used in dentistry. (1 Mark)

Gold has melallic luster because it it is the element of transition element and also use in dentistry due to its shinpess.

Extended Response Questions (ERQs)

The following questions (7 and 8) offered a choice between part **a** and **b**.

Majority of the candidates attempted part 'a' of both question 7 and 8. This shows their interest and in-depth understanding of both concepts, i.e. 'empirical and molecular formula' and 'states of matter'. However, a fewer response in part 'b' reflect that mostly candidates lack confidence and command over the knowledge embedded in the concepts of 'formation of ionic bond' and 'rusting'.

Question 7a:

A compound contains 24% carbon, 12% hydrogen and 64% oxygen by mass. The relative molecular mass of this compound is 50.057 g/mol.

i. Calculate the empirical formula of the given compound.

ii. Determine its molecular formula.

(**Note**: Atomic mass of C = 12 amu, H = 1 amu and O = 16 amu)

Better responses had accurate and systematically written steps needed for the designed calculations of empirical and molecular formulae. Candidates used the correct formulae for calculating the number of moles of each element and in part 'i', candidates clearly indicated the simplest atomic ratios for determining the empirical formula. Furthermore, candidates in part 'ii' worked out the value of 'n' using the correct formula and values to find the molecular formula.

Example:

To find the empirical formula:

	Carbon	Hydrogen	Oxygen
Given mass :	24	12	64
Molecular mass:	12	1	16
mole = Given mass / mol. mass	24 / 12	12 / 1	64 / 16
Number of moles	2	12	4
(simplest ratio)	÷ 1	÷ 6	÷ 2

Empirical Formula : CH_6O_2

Statement :

The Empirical Formula of the given compound is CH_6O_2.

To find Molecular Formula:

Data:

Empirical Formula = CH_6O_2

Molecular Formula = ?

Empirical Formula Mass = ?

Molecular Formula Mass = 50.057 $g\,mol^{-1}$

number = ?

Formula:

$$\frac{\text{Molecular Formula Mass}}{\text{Empirical Formula Mass}} = n$$

Molecular Formula = (Empirical Formula)n.

Solution:

Epirical Formula mass =

$CH_6O_2 = 12\times1 + 1\times6 + 16\times2$

$= 12 + 6 + 32$

$= 50\ g.$

$$n = \frac{50.057}{50}$$

$n = 1.001 \approx 1$

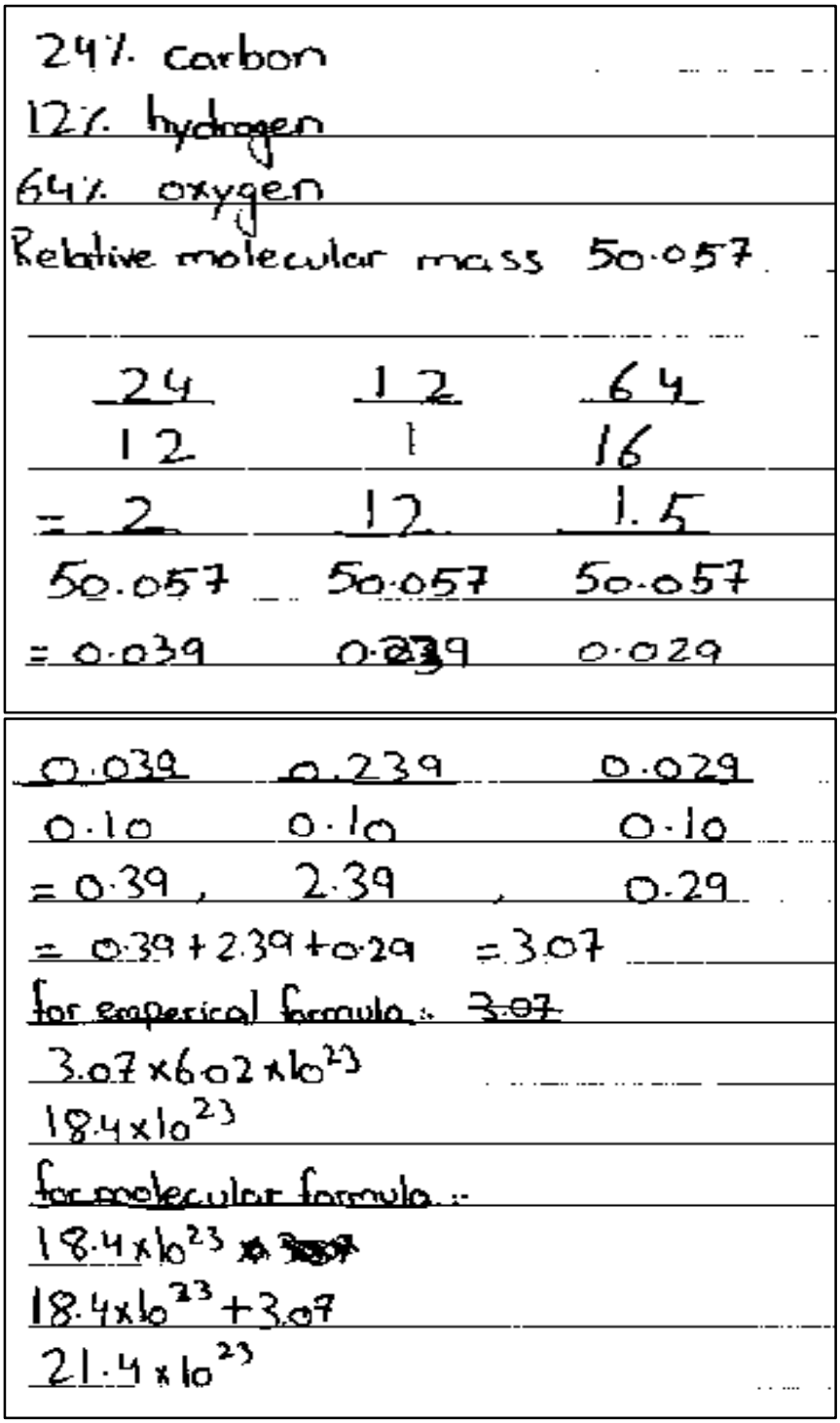
Molecular Formula =

$(CH_6O_2)_1 = CH_6O_2$

The molecular formula of the compound is CH_6O_2.

Weaker responses showed that the candidates were unable to present sequential data/ value substitutions using the correct formula. Majority of the respondents directly used percentages as given in the question for finding out the atomic ratios rather than finding out moles. Most of the candidates used incorrect formulae and substituted wrong values. They were unable to calculate the correct empirical formula which led to the calculation of an incorrect value of 'n' that results the incorrect molecular formula of the given compound.

Example:



Question 7b:

Given below are four elements.

Non-Metal	Metal
Chlorine ($^{35}_{17}Cl$)	Magnesium ($^{24}_{12}Mg$)
Oxygen ($^{16}_{8}O$)	Potassium ($^{39}_{19}K$)

i. Identify TWO elements from the given table which, when form ions, show the same electronic arrangement as a neon ($^{20}_{10}Ne$) atom.

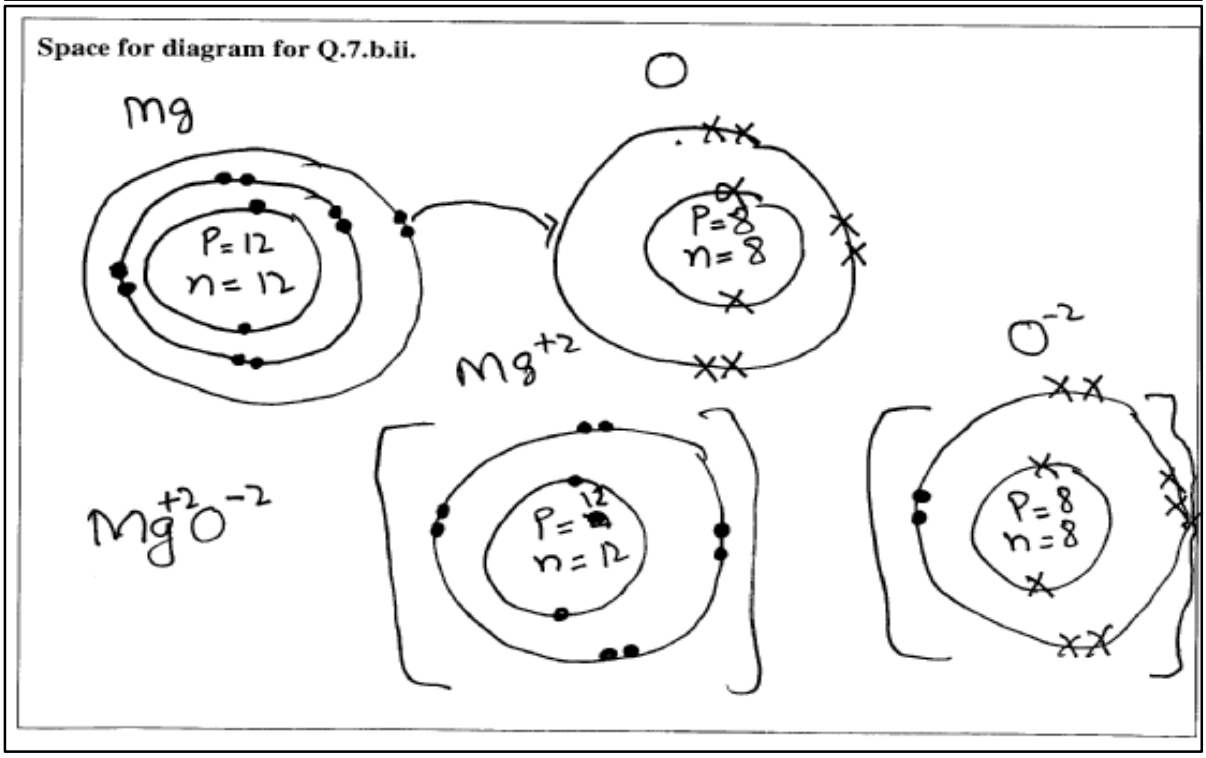
ii. Illustrate the formation of an ionic compound using the elements identified in part i with the help of dot and cross structure of the compound.

Better responses successfully identified the two elements that acquire neon gas electronic configuration when they come in ionic form in part i. These elements were magnesium and oxygen. In part 'ii', these responses remarkably explained/ illustrated all the necessary steps for the formation of ionic compound, i.e. magnesium oxide (MgO). The steps candidates were able to mention included the loss of 2 electrons from the valence shell of magnesium atom results in the formation of Mg^{+2} ion. Likewise, gain of 2 electrons in the valence shell of oxygen atom forms of O^{-2} ion was clearly mentioned. Additionally, these responses had sound clarity of the concept of stability in the transfer of electron from metal to non-metal with the formation of strong electrostatic force of attraction between ions. Many of the better responses explained the formation of ionic bond with the help of chemical equations or electronic configuration of magnesium and oxygen in reference to the electronic configuration of neon atom. Candidates illustrated the formation of MgO using dot and cross electronic structures of atoms with all the necessary labels.

Example:

i. oxygen and Magnesium form ions oxygen needed 2 electrons to attain stability and magnesium has 2 extra electrons, Magnesium will give two electron to oxygen and both elements will attain stability, Magnesium will become positive ion and oxygen with gaining two electrons will become negative ion, These both elements will also attain the electronic arrangement of Neon atom.

Space for diagram for Q.7.b.ii.



Weaker responses failed to identify the correct atoms in part 'i'. Candidates did not compare the electronic configuration of ions with the electronic configuration of neon atom. Most of the candidates predicted potassium and chlorine as the answer to part 'i'. In part ii, many of the responses failed to explain the loss or gain of correct number of electrons from the respective atoms. Candidates showed lack of knowledge regarding transferring or sharing of electrons in the formation of ionic compounds. These responses mostly wrote about sharing of electrons between atoms with the formation of a single covalent bond. Such responses failed to illustrate the ionic bond formation through dot and cross structure.

Example:

(b) (i) Chlorine and potassium form ions and the electronic arrangement of both is chlorine ($^{35}_{17}Cl$) and potassium ($^{39}_{19}K$).

(ii) Chlorine :- 17

$1s^2, 2s^2p^6, 3s^2p^5$

(ii) Potassium :- 19

$1s^2, 2s^2p^6, 3s^2p^6, 4s^1$

(iii) Chlorine loss its 7 electrons and potassium also loss its 1^2 electrons. These both elements loss its electrons and make itself stable.

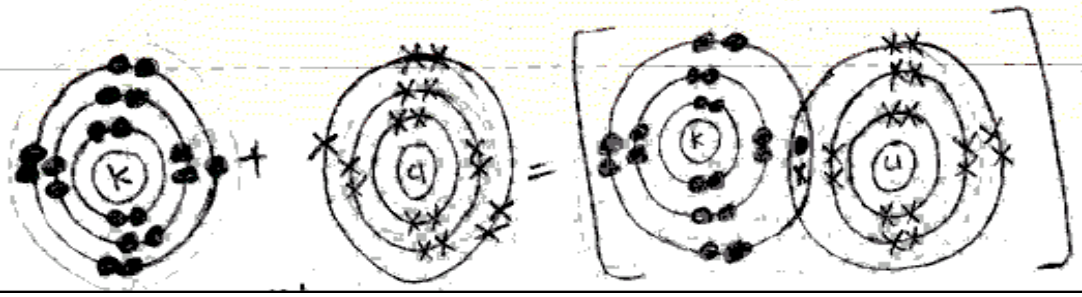
(i) The elements which forms ions are chlorine and potassium when Potassium chloride ~~dissolved~~ dissolved in water, it ionizes completely and form many ions and are also good conductors of electricity in solutions.

(ii) Formation of an Ionic Compound :-

→ Ionic compounds are formed when an atom of a molecule completely transfers it's electrones to the other atom of a molecule. Usually Group-1 and Group-2 elements are involve in ionic compounds because they have tendecy to lose their electrones and attain a high metallic character. Ionic compounds have high melting and boiling points and are the good conducters of heat and electricity.

Space for diagram for Q.7.b.ii.

Potassium chloride ~~and magnesium oxide~~



Question 8a:

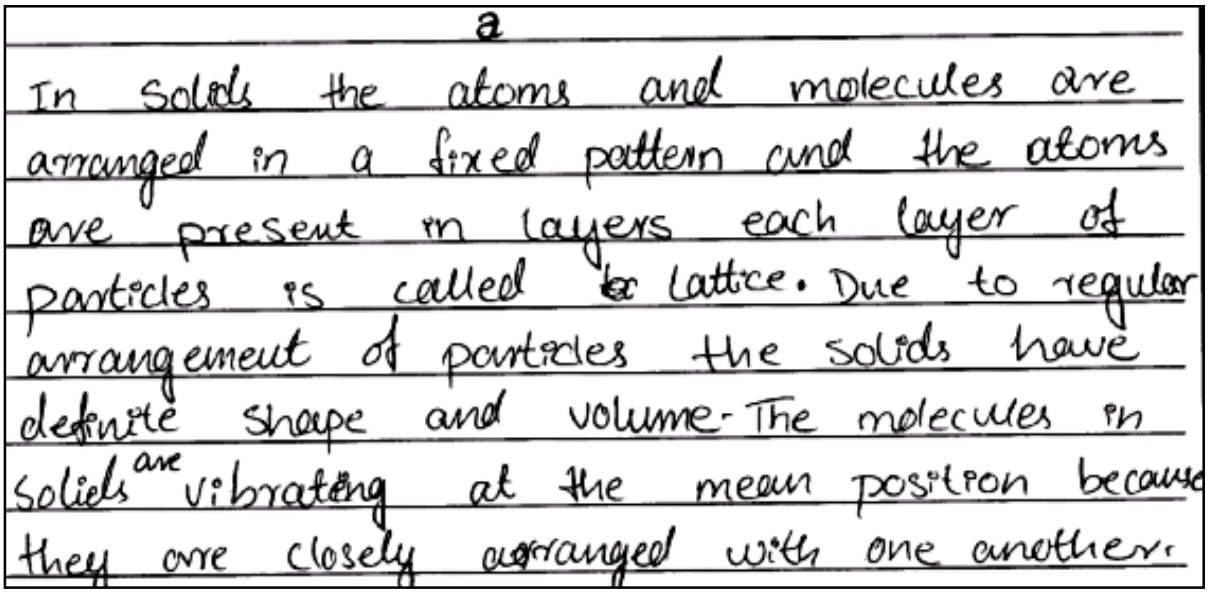
i. Describe the arrangement and movement of particles in solids, liquids and gases.

ii. What happens to the particles during effusion in gases?

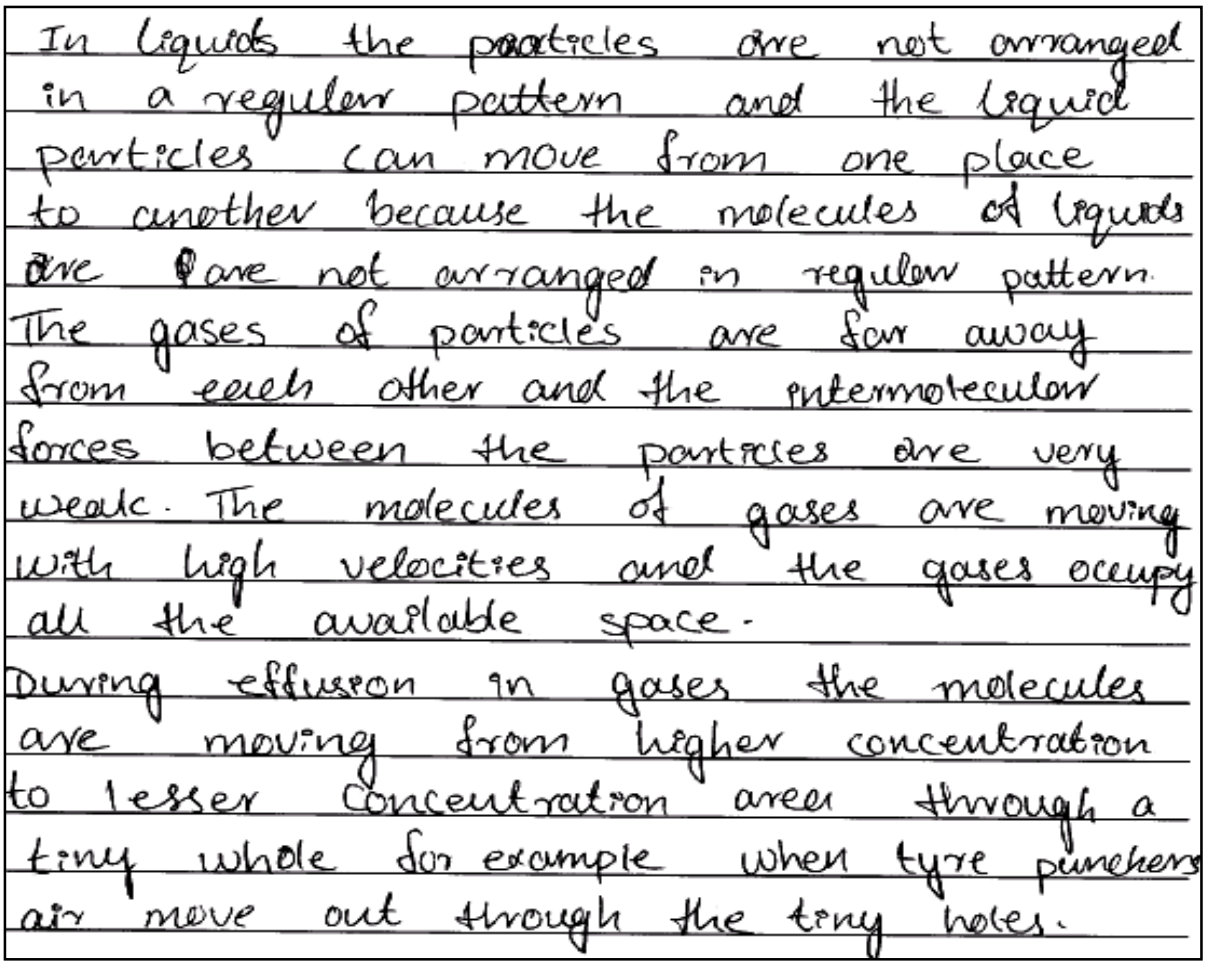
Better responses, in part 'i', described the arrangement of particles in solids liquids and gases, correctly. In these responses, candidates clearly mentioned the closed packing of particles in solids whereas, loose packing of particles in liquids. A few of these responses also mentioned about the clustered arrangement of particles in liquids. In gases, particles randomly occupy the available space as there is no specific arrangement of particles. Moreover, candidates explained the vibrational movement of particles in solids, the slide over movement in liquids and the rapid random movement in gases.

In part ii, candidates were able to elaborate the action of particles of gases during effusion i.e. particles escape one after another through a tiny hole. Many of the responses successfully elaborated the phenomenon of effusion with the example of a punctured tyre.

Example:


a

In Solids the atoms and molecules are arranged in a fixed pattern and the atoms are present in layers each layer of particles is called a lattice. Due to regular arrangement of particles the solids have definite shape and volume. The molecules in solids are vibrating at the mean position because they are closely arranged with one another.



Weaker responses failed to explain the movement of particles in solids, liquids and gases but candidates in many of these responses mentioned the correct arrangement. In part 'i', candidates showed lesser knowledge of the vibrational motion in solids, the slide over movement in liquids and the random speedy movements of gas particles. In part ii, candidates defined diffusion instead of effusion process in gas particles.

Example:

* ARRANGEMENT OF Solids:

They are lightly packed and the movement will be when it will get a movement particles will start to vibrate.

* ARRANGEMENT OF liquids:

They are loosely packed but not as solid and gas. They will started move like as So high for example when H_2O is kept on the table in a glass, So if movement will take place it will somehow start to come out of the glass as the force that will exert on it.

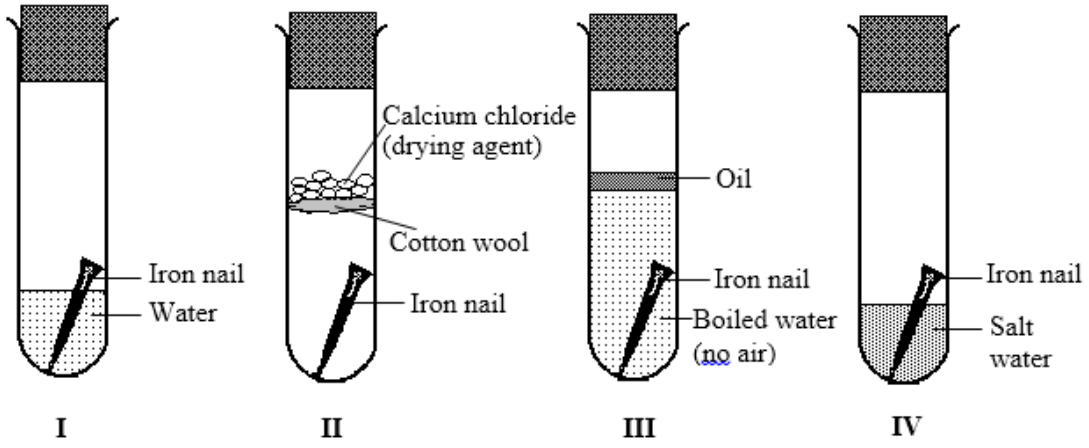
ARRANGEMENT OF Gases:

They are loosely arranged and the movement will be if the pressure will be high so the will push / or interact (and combine) with other molecule and push them in the direction that the pressure in.

(ii) The particles of gases during effusion will be high because of effusion pressure.

Question 8b:

The given diagram shows different conditions to investigate rusting of iron.



i. In which of the given test tubes, **I** to **IV**, will the iron nail rust? Explain why the iron nail rusts in the identified test tube(s) and not in others.

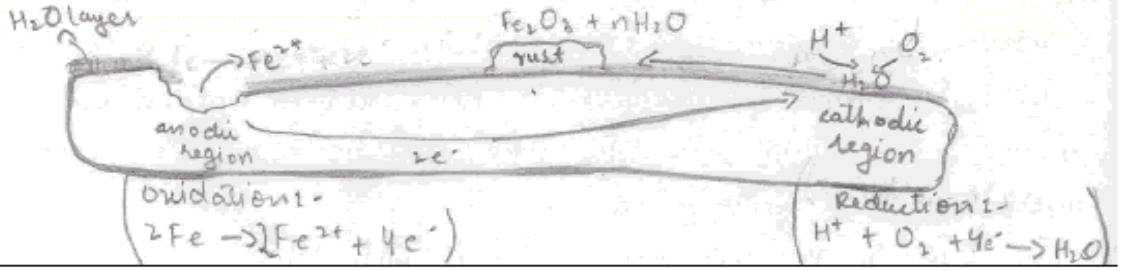
ii. Why is the process of rusting a redox reaction?

Better responses showed that the candidates efficiently identified that test tube number I and IV will show the process of rusting of the iron nail in part 'i'. Furthermore, candidates were able to explain that the test tube I has all the basic requirements for rusting that are water and air which are in contact with the iron nail. In test tube IV, there is salt which is an electrolyte and will accelerate the rusting process. Candidates also wrote the correct reasons for test tube II and III. In test tube II, there is no moisture because calcium carbonate is used as a drying agent that absorbs moisture from the air. In test tube III, there is no air/ dissolved oxygen is absent since the water is boiled and the layer of oil restricts the contact of air with water. In part 'ii', the process of rusting was justified as a redox reaction with respect to iron and oxygen, i.e. iron is oxidised by gaining oxygen/ losing electrons and oxygen is reduced by gaining electrons.

Example:

Corrosion :- continously eating away of metal by its surrounding.
Corrosion is a general word but corrosion of iron is called rusting.
The best condition for rusting is moist air (air that contain water vapours). No Rusting will take place in air free water vapours or dry air (no water vapours present).
Test I :- In this test tube rusting will take place as the required conditions are provided which is moist air.
Test II :- In this test tubes no rusting of iron nail will take place as Calcium chloride is a drying agent so it will remove water vapours from the surrounding and Hence there will be only dry air. The conditions are incomplete air but no moist environment.
Test III :- No rusting will take place as there is a layer of oil which will not allow air to pass. Hence the layer of oil is acting like a protective shield. The conditions are incomplete moist environment but no air.
Test IV :- Rusting will take place at higher speed than test tube I as there is salt which speed up the process of rusting. and there are required conditions also present (moist air).

The process of rusting is a redox reaction as both Oxidation and reduction process take place.
Oxidation:- loss of electron, loss of hydrogen and gain of Oxygen
Reduction :- gain of electron, gain of hydrogen and loss of Oxygen.



Weaker responses contained the correct identification of the test tube I mostly but failed in correct identification of other test tubes. Candidates also wrote the correct reason for the test tube I but lack of understanding was shown in other cases. Many of the candidates only wrote the labelled content for each test tube without explaining the purpose or impact. Some other responses were that this cotton wool absorbs moisture that is why iron nail in test tube II would not rust. A few of responses also mentioned that there is a quantity difference of water in test tube III that makes the difference in the rusting process and in test tube IV salt will produce acid/ basic/ neutral solution which inhibits rusting because only air and moisture are the key factors for rusting. Besides all the good reasoning, there was a prominent mistake observed even in better responses, i.e. the candidates mentioned that the presence of salt produces acid and that acid is responsible for the rapid rusting of the nail. In part 'ii', candidates failed to explain redox reaction in rusting or they just mentioned the generic definition of the term redox reaction.

Example:

